

ex/ Consider following homogeneous-time varying linear system of differential equations where $\underline{A}(t)$ is n -dimensional

$$\underline{x}' = \underline{A}(t) \underline{x} \quad (\text{Eq. 1})$$

where $\underline{A}(t) = \underline{V} \underline{\Lambda}(t) \underline{V}^{-1}$

↖ diagonal

, i.e., for each t , matrix $\underline{A}(t)$ has the same eigenvectors but corresponding eigenvalues are functions of time.

Then we can define:

$$\underline{z} = \underline{V}^{-1} \underline{x}$$

and form the corresponding D.E. by substituting $\underline{x} = \underline{V} \underline{z}$ in (Eq. 1) to get:

$$\underline{V} \underline{z}' = \underline{V} \underline{\Lambda}(t) \underline{V}^{-1} \underline{V} \underline{z}$$

$$\Rightarrow \underline{z}' = \underline{\Lambda}(t) \underline{z} \quad (\text{Eqn. 2})$$

This final form is in the form of n first order linear time-varying differential equations:

$$\underline{z}'_k = \lambda_k(t) \underline{z}_k, \quad 1 \leq k \leq n$$

or $\frac{dz_k}{dt} = \lambda_k(t) z_k \Rightarrow \int_{z_0}^z \frac{dz_k}{z_k} = \int_0^t \lambda_k(t) dt$

$$\Rightarrow \int \ln|z_k| = \int_0^t \lambda_k(\tau) d\tau + d_k$$

$$z_k(t) = C_k \cdot e^{\int_0^t \lambda_k(\tau) d\tau}$$

Therefore, $\underline{x}(t) = \underline{V} \underline{z}(t) = \sum_{k=1}^n \underline{V}_k z_k(t) = \sum_{k=1}^n \underline{V}_k C_k e^{\int_0^t \lambda_k(\tau) d\tau}$

Note that if $\underline{x}(0) = \underline{V}_m$ then $C_k = 0$ for $k \neq m$ and $C_m = 1$

Therefore $\underline{\Phi}(t; 0, \underline{V}) = \left[\underline{V}_1 e^{\int_0^t \lambda_1(\tau) d\tau} \quad \dots \quad \underline{V}_n e^{\int_0^t \lambda_n(\tau) d\tau} \right]$

Fundamental matrix

$$\underline{\underline{\Phi}}(t; 0, \underline{\underline{V}}) = \underline{\underline{V}} \begin{bmatrix} e^{\int_0^t \lambda_1(\tau) d\tau} & & \\ & \ddots & \\ & & e^{\int_0^t \lambda_n(\tau) d\tau} \end{bmatrix}$$

The state transition matrix can be found as:

$$\underline{\underline{\Phi}}(t; 0) = \underline{\underline{\Phi}}(t; 0, \underline{\underline{V}}) \underline{\underline{V}}^{-1} = \underline{\underline{V}} \begin{bmatrix} e^{\int_0^t \lambda_1(\tau) d\tau} & & \\ & \ddots & \\ & & e^{\int_0^t \lambda_n(\tau) d\tau} \end{bmatrix} \underline{\underline{V}}^{-1}$$

$$\text{Hence } \underline{\underline{\Phi}}(t; t_0) = \underline{\underline{\Phi}}(t; 0) \underline{\underline{\Phi}}^{-1}(t_0; 0)$$

$$= \underline{\underline{V}} \begin{bmatrix} e^{\int_0^t \lambda_1(\tau) d\tau} & & \\ & \ddots & \\ & & e^{\int_0^t \lambda_n(\tau) d\tau} \end{bmatrix} \underline{\underline{V}}^{-1} \underline{\underline{V}} \begin{bmatrix} e^{-\int_0^{t_0} \lambda_1(\tau) d\tau} & & \\ & \ddots & \\ & & e^{-\int_0^{t_0} \lambda_n(\tau) d\tau} \end{bmatrix} \underline{\underline{V}}^{-1}$$

$$\underline{\underline{\Phi}}(t; t_0) = \underline{\underline{V}} \begin{bmatrix} e^{\int_{t_0}^t \lambda_1(\tau) d\tau} & & \\ & \ddots & \\ & & e^{\int_{t_0}^t \lambda_n(\tau) d\tau} \end{bmatrix} \underline{\underline{V}}^{-1} \quad (\text{Eqn. 3})$$

ex/ $\underline{\underline{x}}' = \underbrace{\begin{bmatrix} 1-t & \frac{t}{2}-1 \\ 0 & -\frac{t}{2} \end{bmatrix}}_{\underline{\underline{A}}(t)} \underline{\underline{x}}$, find the state-transition matrix.

Since $\underline{\underline{A}}(t)$ is upper-triangular can recognize $\underline{\underline{\lambda}}_1 = (1-t)$ and $\underline{\underline{\lambda}}_2 = -\frac{t}{2}$ as its time-varying eigenvalues.

$$(\underline{\underline{A}} - \underline{\underline{\lambda}}_1 \underline{\underline{I}}) \underline{\underline{V}}_1 = \underline{\underline{0}} \Rightarrow \begin{bmatrix} 0 & \frac{t}{2}-1 \\ 0 & \frac{t}{2}-1 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \underline{\underline{0}} \Rightarrow \overset{\text{can choose}}{\underline{\underline{V}}_1} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$(\underline{\underline{A}} - \underline{\underline{\lambda}}_2 \underline{\underline{I}}) \underline{\underline{V}}_2 = \underline{\underline{0}} \Rightarrow \begin{bmatrix} 1-\frac{t}{2} & \frac{t}{2}-1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \underline{\underline{0}} \Rightarrow \overset{\text{can choose}}{\underline{\underline{V}}_2} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

Therefore :

$$\begin{aligned}
 \underline{A}(t) &= \begin{bmatrix} \underline{V}_1 & \underline{V}_2 \end{bmatrix} \begin{bmatrix} \lambda_1(t) & \\ & \lambda_2(t) \end{bmatrix} \begin{bmatrix} \underline{V}_1 & \underline{V}_2 \end{bmatrix}^{-1} \\
 &= \underbrace{\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}}_{\underline{V}} \underbrace{\begin{bmatrix} 1-t & 0 \\ 0 & -\frac{t}{2} \end{bmatrix}}_{\underline{\Lambda}(t)} \underbrace{\begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix}}_{\underline{V}^{-1}}
 \end{aligned}$$

Then using (Eqn.3) of the example on page 98 :
the state transition matrix can be found as:

$$\underline{\Phi}(t; t_0) = \underline{V} \begin{bmatrix} e^{+\int_{t_0}^t \lambda_1(\tau) d\tau} & 0 \\ 0 & e^{\int_{t_0}^t \lambda_2(\tau) d\tau} \end{bmatrix} \underline{V}^{-1}$$

$$\int_{t_0}^t \lambda_1(\tau) d\tau = \int_{t_0}^t (1-\tau) d\tau = \tau - \frac{1}{2} \tau^2 \Big|_{t_0}^t = (t-t_0) - \frac{1}{2} (t^2 - t_0^2)$$

$$\int_{t_0}^t \lambda_2(\tau) d\tau = \int_{t_0}^t -\frac{\tau}{2} d\tau = -\frac{\tau^2}{4} \Big|_{t_0}^t = -\frac{1}{4} (t^2 - t_0^2)$$

Thus

$$\underline{\Phi}(t; t_0) = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} e^{(t-t_0) - \frac{1}{2}(t^2 - t_0^2)} & 0 \\ 0 & e^{-\frac{1}{4}(t^2 - t_0^2)} \end{bmatrix} \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} e^{(t-t_0) - \frac{1}{2}(t^2 - t_0^2)} & -e^{(t-t_0) - \frac{1}{2}(t^2 - t_0^2)} \\ 0 & e^{-\frac{1}{4}(t^2 - t_0^2)} \end{bmatrix}$$

$$\underline{\Phi}(t; t_0) = \begin{bmatrix} e^{(t-t_0) - \frac{1}{2}(t^2 - t_0^2)} & -e^{(t-t_0) - \frac{1}{2}(t^2 - t_0^2)} + e^{\frac{1}{4}(t^2 - t_0^2)} \\ 0 & e^{\frac{1}{4}(t^2 - t_0^2)} \end{bmatrix}$$